Silverio Martinez Andrés

Nombre:

Tema:

Día	Mes	Año	Folio
01	12	21	

Exámen

1.- Calcule cuanto calor se desprendería si se hacen reacción 325[g] de C_3H_4 con un exceso de O_2 si la reaccion ocurre con un rendimiento del 80%

$$C_3H_{4(g)} + 4O_{2(g)} \rightarrow 3CO_{2(g)} + 2H_2O_{(g)}$$

* Ver si la reaccion esta balanceada

3 C 3

4 H 4 $\quad\therefore$ La reacción está balanceada

8 O 8

* Calcular ΔH_r° a partir de ΔH_f°

$\Delta H_f^\circ\ C_3H_{4(g)} = +185.4 \frac{kJ}{mol}$

$\Delta H_f^\circ\ O_{2(g)} = 0$

$\Delta H_f^\circ\ CO_{2(g)} = -393.5 \frac{kJ}{mol}$

$\Delta H_f^\circ\ H_2O_{(g)} = -241.8 \frac{kJ}{mol}$

Por lo tanto

$$[3mol(\Delta H_f^\circ CO_2) + 2mol(\Delta H_f^\circ H_2O)] - [(\Delta H_f^\circ C_3H_4) + 4mol(\Delta H_f^\circ O_2)]$$

$$\Delta^\circ H_r = [3mol(-393.5 \tfrac{kJ}{mol}) + 2mol(-241.8 \tfrac{kJ}{mol})] - [(185.4 \tfrac{kJ}{mol}) + 4mol(0)]$$

Nombre:

Tema:

Día	Mes	Año	Folio
01	12	21	

$\Delta H^{\circ}_{r} = -1849.5\ kJ$

$325\ g\ de\ C_3H_4 \left(\frac{1\ mol\ de\ C_3H_4}{40.063\ g\ de\ C_3H_4}\right) = 8.112\ mol\ de\ C_3H_4$

Por lo tanto

$\left(\frac{-1849.5\ kJ}{1\ mol\ de\ C_3H_4}\right)\left(8.112\ mol\ de\ C_3H_4\right) = Q'_r = 15003.144\ kJ$

$-\left(1849\ [kJ]\right)\left(\frac{1\ mol}{(40.063)(325)}\right) = \Delta H = 0.1420\ \frac{kJ}{g}$

$0.1420 \times 80 = 11.30$

Rendimiento del 80% $\rightarrow \frac{11.36}{100} = 0.1136\ kJ/g$

Nombre:

Tema:

Día | Mes | Año | Folio

2.-

$N_2O_{5(g)} + H_2O_{(l)} \rightarrow 2HNO_{3(l)}$ $\Delta H_r = -92[kJ]$

$8HNO_{3(l)} \rightarrow 4H_{2(g)} + 12O_{2(g)} + 4N_{2(g)}$ $\Delta H_r = 1392.8[$

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$ $\Delta H_r = -571.66[kJ]$

~~Calc~~ *Comprobar si está balanceada

$2N_{2(g)} + 5O_{2(g)} \rightarrow 2N_2O_{5(g)}$

4 N 4

10 O 10 ∴ Esta balanceada

*Se acomodan e inviertes

$\frac{1}{2} \rightarrow$ $\cancel{2H_{2(g)}} + \cancel{6}^{5}O_{2(g)} + 2N_{2(g)} \rightarrow \cancel{4HNO_{3(l)}}$

$\cancel{2H_2O} \rightarrow \cancel{2H_{2(g)}} + \cancel{O_{2(g)}}$

$2 \rightarrow$ $\cancel{4HNO_{3(l)}} \rightarrow \cancel{2N_2O_{5(g)}} + 2H_2O_{(l)}$

$2N_{2(g)} + 5O_{2(g)} \rightarrow 2N_2O_5$

se suman

$\Delta H_r = (+1392.8[kJ])(-1)\left(\frac{1}{2}\right)$ $\Delta H_r = (-92[kJ])(-1)(2)$

$\Delta H_r = (-571.66[kJ])(-1)$ $\Delta H_r = 59.26[kJ]$

Nombre:

Tema:

Día | Mes | Año | Folio

3.-

$$2NO_{2(g)} \longleftrightarrow N_2O_{4(g)} \quad Kc = 5.6$$

$$Kc = \frac{[N_2O_4]}{[NO_2]^2} = 5.6$$

	$2NO_{2(g)}$	$N_2O_{4(g)}$
inicio	$\frac{0.7M}{7L}$	$\frac{0.42M}{7L}$
cambio	$-2x$	$+x$
equilibrio	$0.1-2x$	$0.06+x$

$$Kc = 5.6 = \frac{[0.06+x]}{[0.1-2x]^2} = \quad x_1 = -0.001224 \quad x_2 = 0.145$$

$$5.6 = \frac{0.06+x}{4x^2-0.4x+0.01}$$

$$[NO_2]_{eq} = 0.1-2x = 0.1-2(-0.001224 = 0.1024[M]$$

$$[N_2O_4]_{eq} = 0.06+x = 0.06+(-0.001224) = 0.058776[M]$$

$$0.06+x = 5.6(4x^2-0.4x+0.01)$$

$$x = 22.4x^2 - 2.24x + 0.056 - 0.06$$

$$x = \overset{a}{22.4}x^2 - \overset{b}{2.24}x - \overset{c}{0.004}$$

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} \qquad x = \frac{-(-2.24) \pm \sqrt{(-2.24)^2-4(22.4)(-0.004)}}{2(22.4)}$$

Nombre:

Tema:

Día | Mes | Año | Folio

7.-

$$CO_{(g)} + Cl_{2(g)} \longleftrightarrow COCl_2$$

$$K_c = 1.5 \times 10^4 = \frac{[COCl_2]}{[CO_{(g)}][Cl_{2(g)}]}$$

$140[dm^3] = 140L$

	$CO_{(g)}$	$Cl_{2(g)}$	$COCl_{2(g)}$
Inicio	$\frac{2.499\,M}{140L}$	$\frac{0.987\,M}{140L}$	0
Cambio	$-x$	$-x$	$+x$
Equilibrio	$0.017 - x$	$0.007 - x$	$0 + x$

$$70_g \text{ de } Cl_2 \left(\frac{1 \text{ mol de } Cl_2}{70.91_g \text{ de } Cl_2} \right) = 0.987 [mol] \text{ de } Cl_2$$

$$70_g \text{ de } CO \left(\frac{1 \text{ mol de } CO}{28.01_g \text{ de } CO} \right) = 2.499 [mol] \text{ de } CO$$

$$K_c = 1.5 \times 10^4 = \frac{[x]}{[0.017 - x][0.007 - x]}$$

0.006953

$$1.5 \times 10^4 = \frac{x}{x^2 - 0.024x + 0.000119}$$

Nombre:

Tema:

Día	Mes	Año	Folio

$$x = 1.5\times10^{4}(x^2 - 0.024x + 0.000119)$$

$$x = \overset{a}{15000}x^2 - \overset{b}{360}x + \overset{c}{1785}$$

$$x = \frac{-(-360) \pm \sqrt{(-360)^2 - 4(15000)(1785)}}{2(15000)}$$

$$x = \frac{360 \pm \sqrt{129600 - 107100000}}{30000}$$

$$x = \frac{360 \pm 10342.649}{30000}$$

$$x_1 = 0.006953 \qquad x_2 = 0.017112$$

$$[CO]_{eq} = 0.017 - x = 0.010047$$

$$[Cl_2]_{eq} = 0.007 - x = 4.7\times10^{-5}$$

$$[COCl_2]_{eq} = x = 0.06953$$